Multi-Modal SLAM Proxy: Fusing 2D LiDAR and Depth Camera for Robust Mapping

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***Abstract*—The report details the design, implementation, and evaluation of a robust autonomous navigation system for mobile robots operating in Global Positioning System (GPS) denied or visually degraded environments. Relying on a single sensor modality for Simultaneous Localization and Mapping (SLAM) introduces a critical point of failure; visual SLAM struggles in low-light conditions, while 2-Dimensional Light Detection and Ranging (LiDAR) provide sparse geometric data without the semantic context needed for advanced tasks. This paper proposes a "Proxy Architecture" that fuses a sparse 2D LiDAR stream with a dense 3D depth camera stream within a Graph SLAM framework. Deployed on a differential drive mobile robot within a Robot Operating System (ROS) 2 and Gazebo simulation environment, the proposed system is evaluated for its resilience during simulated sensor failures. Results demonstrate that the multi-modal fusion approach successfully prevents localization drift during simulated camera blackouts, maintaining precise spatial accuracy.**

Keywords—sensor fusion, ROS2, autonomous navigation, SLAM, RTAB-Map, LiDAR, depth camera

I. Introduction

Autonomous mobile robots (AMRs) are increasingly deployed in logistics, warehouse automation, and unlit indoor environments. A critical requirement for these autonomous platforms is robust, GPS-independent navigation. Traditional approaches often depend on a single primary sensor, such as an RGB-D camera or a 2D LiDAR. However, relying on a single sensor modality creates a single point of failure. Visual SLAM systems fail in the dark, while standard 2D LiDAR's provide sparse geometric data that lacks the dense 3D and semantic context required for identifying specific navigation zones or avoiding complex obstacles. Visual SLAM frameworks such as Oriented FAST and Rotated BRIEF Simultaneous Localization and Mapping (ORB-SLAM2) demonstrate accurate localization using RGB-D sensors but remain sensitive to illumination conditions and texture-poor environments [1]. Conversely, LiDAR-based SLAM systems such as Google Cartographer provide reliable geometric mapping using laser scan data but lack dense visual context for richer environmental understanding [2].

To address these limitations, we hypothesize that fusing a sparse 2D LiDAR stream with a dense 3D Depth Camera stream within a Graph-SLAM framework can create a highly resilient multi-modal architecture. This setup mimics high-end industrial sensor fusion without the cost of a 3D LiDAR. The primary objective of this project is to build and validate a ROS 2 simulation of this sensor-fused system. The specific goal is to configure Real-Time Appearance-Based Mapping (RTAB-Map), a graph-based SLAM framework capable of real-time loop closure detection and multi-session mapping [3], to tightly fuse these sensor inputs and prevent localization drift during a simulated "sensor blackout" (e.g., environmental lighting failure).

II. Related Works

Simultaneous Localization and Mapping (SLAM) is a cornerstone problem in autonomous robotics, enabling a robot to construct a map of an unknown environment while simultaneously estimating its pose within that environment. Over the past two decades, SLAM has evolved from probabilistic filtering approaches to sophisticated optimization-based frameworks that can operate in large-scale, real-time scenarios. The classical formulation of SLAM is inherently a probabilistic estimation problem, where the objective is to estimate the posterior distribution over robot trajectories and map features given noisy sensor observations and control inputs.

Early SLAM approaches were predominantly based on filtering techniques, particularly the Extended Kalman Filter (EKF-SLAM). In EKF-SLAM, the robot poses and map landmarks are jointly estimated under a Gaussian assumption, with linearization applied to handle system non-linearities. While this approach provides a mathematically consistent solution, it suffers from several limitations. The computational complexity grows quadratically with the number of landmarks, making it unsuitable for large environments. Additionally, repeated linearization introduces inconsistency over time, especially in highly dynamic or non-linear scenarios [4]. To address these issues, particle filter-based methods such as Fast SLAM were introduced, decoupling landmark estimation and improving scalability, but still facing challenges in maintaining global consistency.

To overcome the limitations of filtering-based approaches, the SLAM community shifted toward optimization-based methods, particularly graph-based SLAM. In graph SLAM, the problem is formulated as a non-linear least squares optimization problem, where robot poses are represented as nodes and spatial constraints (e.g., odometry, scan matching, loop closures) are represented as edges. The goal is to minimize the overall error across all constraints, thereby producing a globally consistent trajectory. This formulation allows for efficient handling of large datasets and provides a natural mechanism for incorporating loop closure constraints, which are essential for correcting accumulated drift. Optimization frameworks such as g2o (General Graph Optimization) and GTSAM (Georgia Tech Smoothing and Mapping) have significantly advanced the efficiency and scalability of graph-based SLAM systems, making them the foundation of many modern implementations [4].

Visual SLAM has gained widespread attention due to the rich perceptual information provided by camera sensors. These systems typically rely on either feature-based or direct methods. Feature-based approaches, such as ORB-SLAM2, extract distinctive key points (e.g., ORB features) and perform matching across frames to estimate camera motion and detect loop closures [1]. ORB-SLAM2 supports monocular, stereo, and RGB-D configurations and has demonstrated high accuracy in structured indoor environments. Its key strengths include real-time performance, robust loop closure detection, and a well-optimized tracking pipeline. However, visual SLAM systems are inherently sensitive to environmental conditions. Variations in illumination, motion blur, dynamic objects, and low-texture regions can significantly degrade feature extraction and matching performance. In extreme cases, such as poorly lit environments or texture less surfaces, the system may fail to maintain tracking, leading to drift or complete localization failure.

Complementary to vision-based approaches, LiDAR-based SLAM systems rely on precise geometric measurements obtained from laser scanners. These systems construct maps by aligning successive scans using techniques such as Iterative Closest Point (ICP) or scan matching, followed by global optimization through pose graphs. Google Cartographer is a

notable example of a real-time SLAM system that integrates 2D and 3D LiDAR data with loop closure detection and pose graph optimization [2]. LiDAR-based methods are highly robust to lighting variations and can produce accurate geometric representations of the environment. They perform particularly well in structured environments such as corridors and warehouses, where geometric features are prominent. However, LiDAR sensors typically produce sparse point clouds and lack semantic and visual richness. This limitation restricts their ability to capture detailed environmental context and may lead to ambiguities in feature-poor geometric environments.

In addition to standalone visual and LiDAR SLAM systems, Visual-Inertial SLAM (VI-SLAM) has emerged as another important category, integrating camera data with inertial measurements from IMUs. The inclusion of IMU data provides high-frequency motion estimates, improving robustness during rapid motion and reducing drift in feature-scarce environments. Systems such as VINS-Mono and OKVIS demonstrate the effectiveness of tightly coupled visual-inertial fusion. However, these systems still inherit the fundamental limitations of visual sensing and require careful calibration and synchronization between sensors.

Recognizing that no single sensor modality is sufficient to handle all real-world conditions, recent research has increasingly focused on multi-modal SLAM systems that combine complementary sensing modalities. Sensor fusion aims to leverage the strengths of each sensor while mitigating their individual weaknesses. For example, LiDAR provides reliable geometric constraints independent of lighting conditions, while cameras offer dense visual information and semantic cues. When fused effectively, these modalities can significantly improve robustness, accuracy, and resilience to sensor degradation.

RTAB-Map (Real-Time Appearance-Based Mapping) is a widely used graph-based SLAM framework designed for multi-modal sensor fusion and long-term mapping applications [3]. It combines visual loop closure detection with graph optimization and supports a variety of input sources, including RGB-D cameras, LiDAR, and odometry. A key feature of RTAB-Map is its memory management strategy, which enables real-time operation by limiting the number of active nodes while maintaining a global map structure. The framework has been extensively used in both research and practical deployments due to its flexibility and scalability.

Despite the advantages of multi-modal SLAM, several challenges remain. One of the primary challenges is temporal synchronization between heterogeneous sensors operating at different frequencies. Misalignment in sensor timestamps can lead to incorrect data association and degraded mapping performance. Additionally, accurate extrinsic calibration between sensors is critical to ensure consistent spatial alignment. Data fusion strategies must also address issues such as sensor noise, outliers, and conflicting measurements. These challenges make the design and evaluation of multi-modal SLAM systems a complex task, particularly in real-time robotic applications.

Recent studies have demonstrated that multi-modal SLAM systems can significantly outperform single-modality approaches, especially in challenging environments. For instance, when visual tracking fails due to poor lighting, LiDAR data can maintain geometric consistency. Conversely, in environments with sparse geometric features, visual information can provide additional constraints for localization. This complementary behaviour highlights the importance of adaptive sensor fusion strategies that can dynamically adjust the contribution of each modality based on environmental conditions.

Building upon these advancements, the present work investigates a multi-modal SLAM architecture that integrates a 2D LiDAR and an RGB-D depth camera within the RTAB-Map framework. The system is implemented using ROS 2 and evaluated in a Gazebo-based simulation environment that replicates a warehouse scenario. Unlike conventional approaches that treat sensor fusion as a static configuration, this work explores the robustness of different fusion strategies under varying operational conditions, including simulated sensor degradation. The primary objective is to analyse whether multi-modal fusion can improve localization stability, reduce trajectory drift, and maintain mapping performance when individual sensors become unreliable. By focusing on practical deployment challenges such as synchronization, sensor reliability, and real-time constraints, this work contributes toward the development of more robust and adaptable SLAM systems for real-world robotic applications.

III. Methodology and System Design

A. System Overview

The primary goal of this work was to build and evaluate multi-modal sensor fusion architecture for autonomous navigation in a simulated warehouse environment. Rather than relying on single sensor modality, the approach combines complementary sensing technologies to improve the robustness of simultaneous localization and mapping (SLAM) under varying environmental conditions. The entire system was implemented within a Robotic Operating System (ROS) 2 Humble and Gazebo simulation environment using a warehouse world, which allowed repeatable, controlled

testing without the constraints of physical hardware. The overall system architecture used in this work is illustrated in Fig. 1.

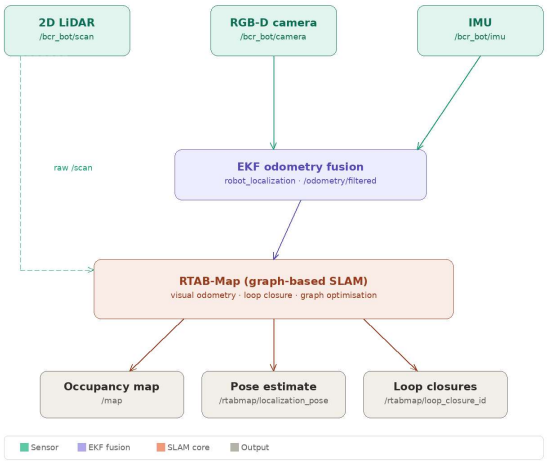

Figure 1. System architecture of the multi-modal SLAM framework.

B. Mobile Robot Platform

The simulated robot platform used throughout this work is the BCR Bot, an open-source differential drive mobile robot developed by Black Coffee Robotics and designed specifically for indoor navigation experiments in ROS 2 and Gazebo [5]. The robot is spawned in the simulation under the namespace bcr_bot, and all sensor data is published under that namespace accordingly.

The BCR Bot uses a two-wheel differential drive mechanism, where independent control of the left and right wheels enables both linear and rotational motion. The translational and rotational velocities of the robot can be derived from the individual wheel angular velocities as follows. Given wheel radius r, wheel separation L, and right and left wheel angular velocities ω_R and ω_L respectively:

$$v = \frac{r(\omega_R + \omega_L)}{2}, \quad \omega = \frac{r(\omega_R - \omega_L)}{L}$$

This kinematic model underpins the wheel odometry estimation fed into the EKF fusion layer described in Section III-D.

C. Boundary Conditions and Assumptions

To keep the scope of the simulation manageable and ensure that results are reproducible, several assumptions were defined at the outset.

The robot operates on a flat, two-dimensional surface throughout all experiments. Vertical displacement along the z-axis is assumed to be zero, and elevation changes are not considered. For the baseline SLAM accuracy evaluation, the warehouse environment is treated as fully static, no moving obstacles were introduced during primary mapping runs. This was a deliberate choice to isolate the effect of sensor fusion from the added complexity of dynamic scene handling. All simulated sensors incorporate Gaussian noise models to approximate the measurement uncertainty present in real hardware, ensuring that results are not overly optimistic relative to physical deployment.

D. Sensor Suite and EKF Fusion

The BCR Bot repository provides a modular sensor configuration controlled through launch arguments. For this work, the depth camera and 2D LiDAR were enabled; the stereo camera was disabled, as RTAB-Map's RGB-D pipeline does not require stereo rectification. The active sensor suite consisted of three modalities:

A 2D LiDAR sensor publishes laser scan data on /scan, capturing the geometric structure of the surrounding environment at high frequency. This forms the primary input for loop closure detection and structural consistency checking. A 3D RGB-D depth camera publishes on /camera/depth/image_raw and /camera/rgb/image_raw, providing dense depth measurements alongside color information used for visual odometry and environmental reconstruction. An Inertial measurement unit, IMU sensor publishes on /imu, measuring linear acceleration and angular velocity at high frequency to provide short-term pose stability between slower sensor updates.

Raw measurements from these sensors are not fed directly into RTAB-Map. Instead, the IMU and wheel odometry sourced from the differential drive controller are first fused using an Extended Kalman Filter (EKF), implemented via the robot localization package. This produces a smoothed filtered odometry estimate on /odometry/filtered, which is passed to RTAB-Map as the motion prior. The raw LiDAR scan is additionally forwarded directly to RTAB-Map for geometric consistency checks and loop closure, bypassing the EKF path. This dual path design was a deliberate architectural decision filtered odometry handles motion estimation, while the raw scan preserves geometric fidelity for place recognition.

E. Localization Framework: RTAB-Map

While the BCR Bot package includes default support for SLAM Toolbox [6], this work uses RTAB-Map (Real-Time Appearance-Based Mapping) instead, as it natively supports heterogeneous sensor fusion combining LiDAR and RGB-D data within a single unified graph-SLAM framework [3]. This made it the most suitable choice for evaluating the multi-modal fusion objective of this project.

Within the configured pipeline, RTAB-Map uses visual odometry derived from the RGB-D stream for frame-to-frame motion estimation, while the LiDAR scan data is used to enforce geometric consistency and trigger loop closures when previously visited areas are detected. When a loop closure is accepted, the system performs graph optimization to correct accumulated drift across the entire map. This complementary arrangement is particularly valuable in scenarios where one modality degrades for example, under low-lighting conditions where visual features become sparse, the LiDAR continues to provide stable geometric constraints, preserving map quality. The experiments in this work explicitly test this failure-mode behaviour, and the results are discussed in Section VI.

F. Functional Diagram and Data Flow

The system data flow follows a layered architecture designed to improve robustness through sensor redundancy. The BCR Bot publishes sensor measurements under the /bcr_bot namespace, including laser scans from the LiDAR (/bcr_bot/scan), RGB-D images from the depth camera (/bcr_bot/camera/depth/image_raw) and (/bcr_bot/camera/rgb/image_raw), and inertial measurements from the IMU (/bcr_bot/imu). Wheel encoder data from the

differential drive controller is published as odometry on (/bcr_bot/odom).

Wheel odometry and IMU measurements are fused using an Extended Kalman Filter implemented through the robot localization package. The EKF produces a filtered pose estimate on /odometry/filtered, which serves as the motion prior for the SLAM system.

RTAB-Map subscribes to the RGB-D camera streams together with the filtered odometry estimate to perform visual odometry and frame-to-frame pose estimation. In parallel, the raw LiDAR scan data is also provided to RTAB-Map to support geometric registration and loop closure detection. When a previously visited location is recognized, RTAB-Map performs pose graph optimization to correct accumulated drift and update the global map.

All sensor data is transformed into a common coordinate frame using the ROS tf2 transformation tree maintained by robot_state_publisher. The primary coordinate frames include base_link for the robot body, laser_frame for the LiDAR sensor, and camera_link for the RGB-D camera. The resulting optimized pose graph is used to generate a globally consistent occupancy grid map of the warehouse environment. The data pipeline and ROS topic integration used in the proposed system are illustrated in Fig. 2.

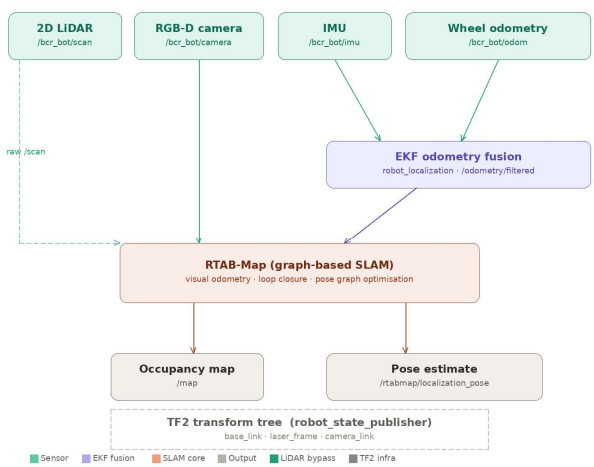

Figure 2. Data flow and topic integration in the proposed multi-modal SLAM framework.

G. Evaluation Methodology

To rigorously assess the proposed multi-modal architecture against single-sensor baselines, a structured quantitative evaluation was conducted. Qualitative map visualisations provide intuition regarding system behaviour, but a rigorous academic evaluation requires quantifiable evidence of trajectory accuracy, mapping stability, and algorithmic robustness under degraded conditions.

To evaluate SLAM accuracy, a perfectly accurate reference trajectory is required. A deterministic baseline was established by extracting the exact robot pose at each simulation timestep directly from the Gazebo physics engine via the /gazebo/modelstates topic.

This ground truth reference trajectory was recorded at 100 Hz. Because the RTAB-Map SLAM node publishes its estimated pose at 10 Hz, the ground truth data was subsequently down-sampled to 10 Hz to ensure exact temporal alignment. Quantitative metrics were computed offline using the evo trajectory evaluation toolkit applied to the recorded rosbags.

The system's performance was measured using three standard trajectory and mapping metrics. Each metric was deliberately chosen to isolate a specific failure mode of autonomous navigation:

1) Absolute Trajectory Error (ATE):

ATE evaluates the global consistency of the map. It measures the accumulated drift over the entire run by comparing the estimated trajectory against the ground truth. It is computed as the Root Mean Square Error (RMSE) of the translation component after performing an optimal rigid-body alignment (Horn's method). Formally, ATE is defined as:

$$ATE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|\mathrm{trans}(Q_i^{-1}P_i)|^2}$$

where P_i is the estimated pose at time i, Q_i is the ground truth pose, Q_i^{-1} is its inverse, and N is the total number of poses. *Why it is used:* ATE proves whether the final generated map is geometrically accurate enough for warehouse operations.

2) Relative Pose Error (RPE):

Unlike ATE, RPE measures local accuracy and incremental drift over short, fixed time windows $(\Delta = 1\ second)$. RPE is calculated as:

$$RPE = \sqrt{\frac{1}{N}\sum_{i}|trans(E_i)|^2}$$

$$\text{where} \quad E_i = (Q_{i+\Delta}^{-1} \cdot Q_i)^{-1} \cdot (P_{i+\Delta}^{-1} \cdot P_i)$$

RPE is highly sensitive to short-term odometry drift and momentary sensor failures. It proves whether the robot's motion is smooth frame-to-frame, which is critical for local collision avoidance.

3) Loop Closure Success Rate (LCSR):

LCSR measures the proportion of spatially valid loop closure detections relative to total detections, distinguishing true positives (TP) from false positives (FP). By cross-referencing the RTAB-Map event log against the ground truth, a closure was classified as a true positive if the estimated distance between matched poses was within 0.5 m of the actual ground truth distance. The rate is defined as:

$$LCSR = \frac{TP}{TP + FP} * 100$$

LCSR evaluates the efficiency of the Graph-SLAM optimiser, exposing vulnerabilities like "perceptual aliasing" where the robot confuses two identical-looking areas.

IV. Implementation

A. Simulation Setup

The system was implemented using ROS 2 Humble operating alongside the Gazebo simulation engine as depicted in Fig. 3. We spawned the robot inside a simulated small warehouse environment. The simulation packages the necessary robot descriptions (URDF) and action graphs required to publish all sensor topics.

The simulation environment warehouse world and the robot's sensor suite were instantiated using custom ROS 2

launch files. The primary sensors published to the ROS 2 network include:

- **2D LiDAR:** Published to the /scan topic, providing planar geometric constraints.
- **RGB-D Camera:** Published to /camera/depth/image_raw and /camera/image_raw, providing dense 3D point clouds and visual features.
- **IMU:** Published to /imu/data, providing high-frequency angular velocity and linear acceleration.

To bridge the physical simulation with the ROS 2 ecosystem, gazebo ros packages plugins were utilized to broadcast joint states and simulate physics.

```
ekf_filter_node:
    ros__parameters:
      frequency: 30.0
      two_d_mode: true
      publish_tf: true

      # Wheel Odometry (Trusting X, Y, and Yaw rate slightly)
      odom0: /bcr_bot/odom
      odom0_config: [true,  true,  false,
                     false, false, true,
                     true,  false, false,
                     false, false, true,
                     false, false, false]

      # IMU Data (Trusting Yaw and Yaw velocity heavily)
      imu0: /imu/data
      imu0_config: [false, false, false,
                    false, false, true,
                    false, false, false,
                    false, false, true,
                    true,  false, false]
```

- **Physical Constraints:** The robot's footprint, inertial tensors, and wheel friction coefficients (μ_1, μ_2) were defined in the URDF to closely mimic real-world slippage during rotational acceleration.
- **Sensor Noise Modeling:** To prevent the SLAM algorithm from over-fitting to perfect simulation data, Gaussian noise models were injected into the Gazebo sensor plugins. For the 2D LiDAR, the measured distance z_k at time step k is modelled as:

$$Zk = dk + \nu k$$

Where d_k is the true geometric distance to the obstacle, and ν_k represents the zero-mean Gaussian measurement noise, with sigma set to 0.01 meters to simulate typical industrial LiDAR variance.

B. *Transform Tree and Odometry Pipeline*

A critical component of the ROS 2 implementation is the maintenance of a rigorous Transform (TF) tree. The structural hierarchy is defined as map -> odom -> base_link -> [sensor_frames]. This hierarchical organization ensures consistent spatial relationships between coordinate frames and enables accurate transformation of sensor data across the system.

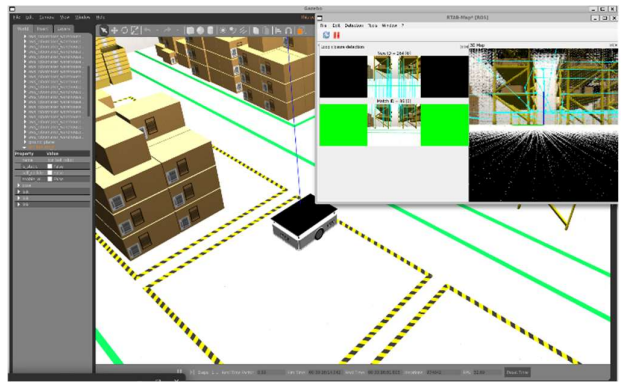

Figure 3: Gazebo simulation environment showing the warehouse setup with the mobile robot navigating between structured obstacles, alongside visualization displaying RGB-D inputs, feature extraction, and point cloud generation used for localization.

Raw wheel odometry inherently accumulates drift due to simulated wheel slip during rotational maneuvers. To mitigate this and stabilize the odom -> base_link transform, the robot localization package was integrated. An Extended Kalman Filter (EKF) node fused the continuous wheel encoder data (/odom) with the IMU gyroscope data (/imu/data) [7]. By combining complementary sensor modalities, the EKF provides a more robust and noise-resilient estimate of the robot's pose compared to relying on a single sensor source.

The Boolean matrix configuration used to pass these variables into the EKF state vector is shown in the snippet below. This matrix explicitly defines which state variables are updated by each sensor input, allowing selective fusion of reliable measurements. We explicitly configure the filter to trust wheel encoders for planar translation (X, Y) while relying entirely on the IMU for rotational heading (Yaw and Yaw Velocity). This design choice minimizes rotational drift and improves heading consistency during navigation.

C. *SLAM Time Synchronization*

A fundamental challenge in heterogeneous sensor fusion is the temporal misalignment of incoming data streams. The differential drive platform utilizes an RGB-D depth camera operating at 30 Hz alongside a 2D LiDAR operating at 10 Hz. In a standard Graph-SLAM pipeline, the optimizer requires strictly synchronized sensor messages to accurately construct temporal nodes. Unsynchronized inputs result in continuous frame dropping, leading to severe map fragmentation and tracking failure. This mismatch in sensor frequencies introduces temporal inconsistencies that degrade the reliability of feature association across consecutive frames.

To resolve this temporal disparity, the rtabmap package was integrated into the ROS 2 architecture [8]. Specifically, the sync node acts as an intermediary buffer, employing an Approximate Time synchronization policy. This policy aligns the header timestamps of the RGB, depth, camera intrinsics, and IMU streams within a defined temporal slop window, publishing a unified, synchronized rgbd_image topic to the core SLAM node. This synchronization strategy ensures that multi-modal sensor inputs are temporally coherent, thereby improving the stability of graph optimization and reducing the likelihood of tracking failure.

The architectural configuration defining this synchronization and the subsequent multi-modal Graph-SLAM optimization is provided in the launch snippet below:

```
# RTAB-Map Synchronization Node Configuration
Node(
    package='rtabmap_sync',
    executable='rgbd_sync',
    name='rgbd_sync',
    parameters=[{
        'use_sim_time': True,
        'approx_sync': True
    }],
    remappings=[
        ('rgb/image',       '/bcr_bot/kinect_camera/image_raw'),
        ('depth/image',
'/bcr_bot/kinect_camera/depth/image_raw'),
        ('rgb/camera_info',
'/bcr_bot/kinect_camera/camera_info'),
        ('imu',             '/bcr_bot/imu')
    ]
),
# Core SLAM Node Optimization Constraints
Node(
    package='rtabmap_slam',
    executable='rtabmap',
    name='rtabmap',
    parameters=[{
        'use_sim_time': True,
        'frame_id': 'base_link',
        'subscribe_rgbd': True,
        'subscribe_scan': True,
        'subscribe_imu': True,

        # Algorithmic and Kinematic Constraints
        'Reg/Strategy': 1,
        'Reg/Force3DoF': True,
        'Icp/VoxelSize': 0.05,
    }],
    remappings=[
        ('rgbd_image', '/rgbd_image'),
        ('scan', '/bcr_bot/scan'),
        ('odom', '/bcr_bot/odom')
    ]
)
```

Algorithmic Impacts and Boundary Conditions:

The robust performance of the "Proxy Architecture" relies on the explicit declaration of several boundary conditions and optimization strategies within the core node:

- **approx_sync (Temporal Slop):** Set to True, this variable mathematically relaxes the requirement for identical message timestamps. It prevents the optimizer from discarding the disparate 30 Hz and 10 Hz inputs, enabling continuous multi-modal node generation.
- **Reg/Strategy (Fusion Policy):** Initialized to *1*, this parameter dictates the core registration algorithm. It mandates a hybrid approach, executing Iterative Closest Point (ICP) scan matching on the 2D LiDAR data for global geometric consistency, while utilizing visual odometry for dense, local frame-to-frame tracking.
- **Reg/Force3DoF (Kinematic Constraint):** Set to True, this enforces a critical boundary condition. Because the bot is a differential drive ground vehicle, its physical motion is strictly planar. These variables forces the SLAM optimizer to ignore calculated drift in the z-axis (translation) as well as roll and pitch (rotation), preventing erroneous vertical map deviations.
- **Icp/VoxelSize (Computational Impact):** Defines the spatial resolution for the LiDAR point cloud downsampling. A value of *0.05* m was selected to optimize the computational load (CPU utilization) without compromising the geometric accuracy required for precise loop closures.

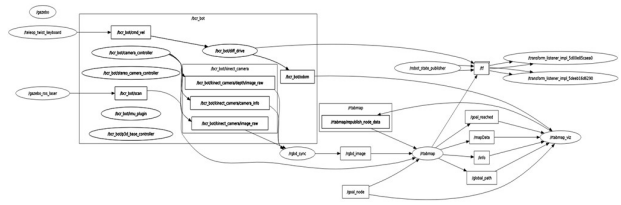

Figure 4: ROS 2 rqt graph illustrating the RTAB-Map sensor fusion pipeline, showing data flow between RGB-D camera topics, LiDAR inputs, odometry estimation, and the RTAB-Map node for mapping and localization.

D. Navigation Stack Integration

To validate the SLAM outputs, the generated map was integrated with the ROS 2 Navigation 2 (Nav2) stack. RTAB-Map projects the dense 3D point cloud and 2D LiDAR data into a 2D occupancy grid published on the /map topic.

The Nav2 stack consumes this grid to populate its global and local cost maps. The local cost map was configured with a rolling window utilizing the synchronized sensor streams to dynamically route the robot around unmapped obstacles. By successfully sending Nav2 *MapsToPose* action goals, we validated that the multi-modal SLAM outputs were mathematically consistent and practically usable for autonomous warehouse traversal.

E. Graph Optimization Model

The core of the multi-modal data processing within the RTAB-Map "Proxy Architecture" relies on Graph-SLAM optimization. Instead of relying on a single filter, the system constructs a pose graph where nodes represent the robot's state over time, and edges represent the sensory constraints. To achieve multi-modal fusion, the algorithm seeks to find the optimal trajectory, defined as the sequence of poses x*, by minimizing the joint non-linear least-squares cost function of all active sensor modalities. The objective function is formulated as:

Variable Impacts and Processing Logic:

- x*: The optimized 2D trajectory path (resolving the drift).
- $e_{vo,i}$, $e_{icp,j}$, $e_{odom,k}$: These represent the residual error vectors for Visual Odometry (Depth Camera), Iterative Closest Point (2D LiDAR), and the EKF Odometry prior, respectively.

- $\sum^{-1}$: The inverse covariance (information matrix) associated with each sensor measurement. This acts as the "trust" weight.

$$x^* = argmin_x \left(\sum_{\{i \in C_{\{vo\}}\}} e^T_{\{vo,i\}} \Sigma_{\{vo,i\}} {}^{\{-1\}} e_{\{vo,i\}} + \sum_{\{j \in C_{\{icp\}}\}} e^T_{\{icp,j\}} \Sigma_{\{icp,j\}} {}^{\{-1\}} e_{\{icp,j\}} + \sum_{\{k \in C_{\{odom\}}\}} e^T_{\{odom,k\}} \Sigma_{\{odom,k\}} {}^{\{-1\}} e_{\{odom,k\}} \right)$$

During the "blackout tests" when the lighting fails, the visual features disappear, causing the visual covariance vo to approach infinity. Mathematically, the inverse covariance $\sum_{vo}$: approaches zero, effectively canceling out the visual error term $\sum_{vo}^{-1}$: in the objective function. The optimizer dynamically and automatically shifts the mathematical weight entirely to the LiDAR (e_{icp}) and EKF (e_{odom}) constraints, ensuring the trajectory x* remains stable without catastrophic divergence.

F. Implementation of Metric Computation:

For each experimental configuration, the robot was driven along the same predefined closed-loop trajectory to ensure consistency across runs. During execution, both the estimated trajectory from RTAB-Map (/rtabmap/odom) and the ground truth trajectory obtained from the Gazebo simulator (/gazebo/model_states) were recorded using ROS 2 bag files.

The recorded data was processed offline using the evo trajectory evaluation toolkit. The ground truth trajectory was first down-sampled to match the frequency of the SLAM output (10 Hz), ensuring temporal alignment. Both trajectories were then spatially aligned using rigid-body transformation before computing error metrics.

Absolute Trajectory Error (ATE) was obtained directly as the RMSE output from the evo_ape tool, representing global drift over the entire trajectory. Relative Pose Error (RPE) was computed using evo_rpe with a fixed time interval (Δ = 1 second) to capture local motion inconsistency and short-term drift.

Loop Closure Success Rate (LCSR) was determined by analyzing the RTAB-Map database and loop closure events. Detected loop closures were cross validated against the ground truth trajectory and classified as valid if the spatial deviation between matched poses remained within a predefined threshold. The ratio of valid detections to total loop closure attempts was then computed.

This pipeline was applied uniformly across all sensor configurations to ensure a fair and reproducible comparison of SLAM performance.

V. EXPERIMENTAL SETUP

A. Simulation Environment

The performance of the proposed multi-modal SLAM architecture was evaluated in a simulated environment using ROS 2 Humble and the Gazebo simulation platform. The robot platform used for all experiments was the BCR Bot differential drive robot, which was deployed inside a warehouse simulation environment.

The simulation environment provides a controlled setting for testing localization and mapping algorithms while ensuring reproducibility across multiple experimental runs. Gazebo physics plugins were used to simulate realistic robot dynamics, including wheel friction, inertia, and sensor noise. The robot was controlled through standard ROS 2 nodes that published sensor data and odometry information to the ROS network.

The primary objective of the experiments was to evaluate the robustness of the proposed multi-modal SLAM architecture under different sensor configurations and to observe how the system behaves when individual sensors become unavailable.

B. Sensor Configuration

The simulated BCR Bot platform was equipped with a heterogeneous sensor suite consisting of a 2D LiDAR sensor, an RGB-D depth camera, and an Inertial Measurement Unit (IMU).

The 2D LiDAR sensor produced planar laser scan measurements and published range data to the /scan topic. These measurements provide geometric information about surrounding obstacles and are primarily used for scan matching and loop closure detection within the SLAM framework.

The RGB-D depth camera generated synchronized color and depth images, which were published to /camera /rgb/image_raw and /camera/depth/image_raw. These images allow RTAB-Map to extract visual features and estimate frame-to-frame motion using visual odometry.

The IMU provided high-frequency measurements of angular velocity and linear acceleration, published to /imu. These measurements were fused with wheel encoder data using an Extended Kalman Filter (EKF) implemented through the robot localization package. EKF produced a filtered odometry estimate on /odometry/filtered, which served as the motion prior for the SLAM system.

The fusion of wheel encoder data and IMU measurements significantly improve short-term pose estimation and reduce odometry drift during rotational motion.

C. Dynamic Lighting Manipulation

To physically simulate the "sensor blackout" during the experimental runs, the global illumination within the Gazebo small warehouse world was dynamically manipulated. Midway through the robot's navigation task, the simulated environmental light source (the directional sun model) was disabled via Gazebo's lighting properties. This approach effectively plunged the warehouse into darkness, naturally reducing the RGB-D camera's visual feature extraction to zero. This method provides a more realistic hardware failure simulation than simply terminating the camera's ROS 2 node, as the SLAM algorithm must organically detect and handle the sudden loss of visual confidence.

VI. Results and Evaluation

A. Sensor Analysis

This section presents the experimental results and evaluates the performance of the proposed SLAM system under different sensor configurations and operating conditions. The robot was tested in multiple scenarios, including normal operation, reduced lighting, and sensor failure cases, to analyse the robustness of each configuration. First, the system behaviour is qualitatively assessed through trajectory and map observations. This is followed by a quantitative analysis using standard evaluation metrics to compare accuracy, stability, and reliability across all configurations.

1) Camera-Only Performance

In the camera-only configuration, the robot relied entirely on its RGB-D camera to understand where it was and build a map of the surrounding environment. The camera works by detecting and tracking visual features — edges, textures, and surface patterns — and using changes in those features between frames to estimate movement. The experiment was conducted under normal lighting initially, followed by a transition to complete darkness to simulate realistic warehouse conditions.

As shown in Fig. 5, the system resulted in a near-total mapping failure. The trajectory is confined to a small region near the entry corridor, while the rest of the environment remains unmapped, indicating that tracking was lost early and could not be recovered. During the initial phase, the system was able to extract sufficient visual features to construct a limited local map. However, as the robot moved into the warehouse interior, the lack of distinctive features and the presence of repetitive structures led to increasing instability in pose estimation and gradual drift.

When the lighting was removed, performance degraded completely. The camera could no longer extract reliable features, causing visual odometry to fail and preventing any loop closure. With no alternative sensing modality available, the system lost localization entirely.

Overall, this configuration highlights a critical limitation of vision-only SLAM: it is highly dependent on both environmental texture and lighting conditions, making it unreliable for deployment in large, feature-poor, and dynamically lit warehouse environments.

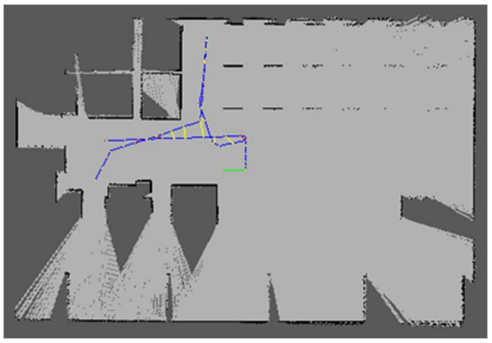

Figure 5: RTAB-Map visualization of camera-only SLAM showing the reconstructed map and estimated robot trajectory based solely on RGB-D visual odometry with lights on.

2) LiDAR-Only Performance

In the LiDAR-only configuration, the robot relied exclusively on laser-based sensing to build and navigate the warehouse map. The system successfully traced the entire boundary of the environment, and the estimated trajectory confirms that the robot completed its full path without losing localization. Furthermore, loop closure was achieved, as the system repeatedly identified previously visited locations and established constraints to maintain global consistency.

However, the results reveal a more complex behaviour in the underlying SLAM process. The map is dominated by a dense web of loop closure links, forming an almost continuous mesh across the environment. In a well-functioning system, such constraints should be sparse and occur only when the robot genuinely revisits a location. In this case, their excessive density indicates that loop closure detection was triggered far too frequently.

This behaviour arises from the inherent limitations of geometry-only sensing. In structured environments such as warehouses, many regions (e.g., aisles) appear geometrically identical to a 2D LiDAR sensor. Without complementary visual information, the system cannot reliably distinguish between similar locations. As a result, the algorithm continuously attempts to match incoming scans with previously stored observations, leading to a large number of redundant and potentially incorrect loop closure constraints.

Consequently, the Graph-SLAM optimizer becomes saturated, increasing computational load and introducing the risk of false-positive loop closures. While the final map appears complete, the underlying optimization process is inefficient and unstable. This highlights a key limitation of LiDAR-only SLAM: although it can reconstruct the environment, it struggles in perceptually ambiguous settings, where reliance on geometry alone can degrade robustness and scalability.

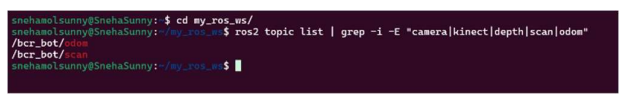

Figure 6: ROS 2 topic inspection showing active /scan and /odom topics, confirming LiDAR-only SLAM configuration with no camera or depth data streams

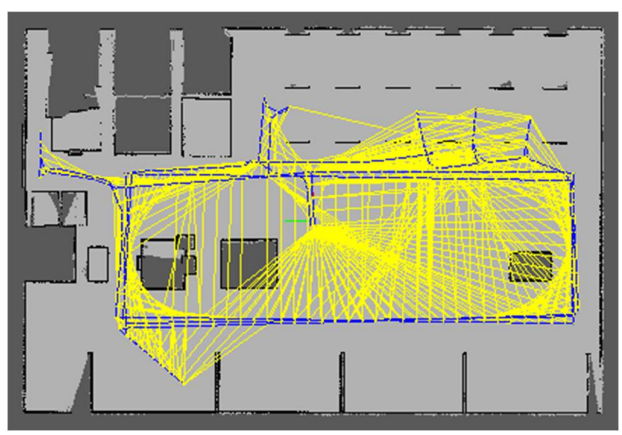

Figure 7: RTAB-Map output illustrating LiDAR-based SLAM results, including the occupancy grid map and pose graph with node connectivity representing the robot trajectory

3) Sensor- Fused Performance

In the multi-modal configuration, both the 2D LiDAR and RGB-D camera were enabled, allowing RTAB-Map to utilize both geometric and visual information for mapping and localization.

The 2D occupancy grid generated accurately represents the warehouse layout, and the estimated robot trajectory closely follows the actual path with minimal deviation. As illustrated in Fig. 6, the trajectory is highly orthogonal, precisely aligning with the warehouse aisles a direct result of the LiDAR sensor providing reliable, long-range geometric scan-matching constraints.

At the same time, the RGB-D camera contributed visual features that improved short-term, frame-to-frame pose estimation, particularly in regions where geometric matching alone may be less distinctive, such as repetitive aisle structures.

This complementary use of sensors enabled effective loop closure detection when the robot revisited previously mapped locations. The resulting loop closure constraints were incorporated into the pose graph to correct accumulated drift and maintain overall map consistency. Compared to the LiDAR-only configuration, the pose graph shows more controlled and meaningful loop closure connections, avoiding excessive or redundant constraints.

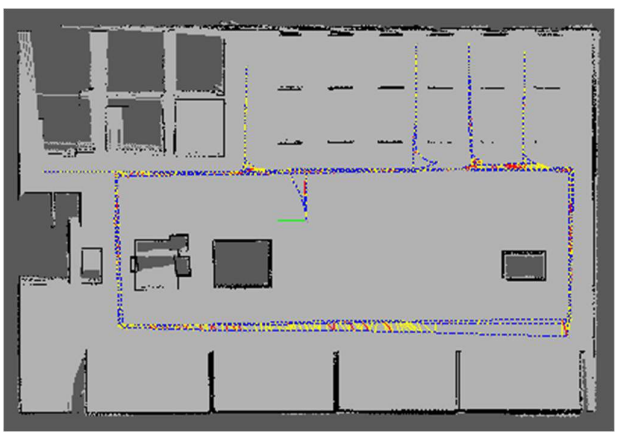

Figure 8: RTAB-Map visualization of multi-modal SLAM using LiDAR and RGB-D sensors, showing a consistent occupancy grid map and optimized robot trajectory with loop closure correction

B. Evaluation Analysis

The experiments demonstrate that multi-modal sensor fusion significantly improves the robustness of the SLAM system. By combining LiDAR and RGB-D data within the RTAB-Map framework, the system benefits from the complementary strengths of both sensing modalities.

The LiDAR sensor provides stable geometric constraints that remain reliable under poor lighting conditions, while the RGB-D camera contributes dense visual information that improves motion estimation and environmental representation.

As a result, the multi-modal configuration achieved the most reliable mapping performance and minimized localization drift compared to the single-sensor configurations.

While the static map comparisons validate the baseline capabilities of each sensor, the dynamic blackout test serves as the definitive proof of the proposed "Proxy Architecture's" resilience. During this specific experimental scenario, the robot-initiated mapping utilizing the full multi-modal fusion architecture under optimal lighting.

Midway through the trajectory, the Gazebo environment lighting was completely disabled. As anticipated, the RGB-D visual odometry pipeline immediately failed to extract and match visual features due to the sudden darkness, issuing tracking loss warnings. However, because the RTAB-Map core was configured with a hybrid registration strategy, the Graph-SLAM optimizer seamlessly shifted its localization weight entirely to the Iterative Closest Point (ICP) scan matching provided by the active 2D LiDAR.

The system maintained its spatial localization without any catastrophic coordinate jumps, odometry drift, or map fragmentation. The final output of this dynamic test is reflected in the fused trajectory (previously shown in Fig. 6), which demonstrates a continuous, highly stable path that perfectly preserved map integrity despite the severe, instantaneous visual degradation.

The evaluated sensor configurations included: (i) Multi-Modal Fusion (LiDAR + RGB-D + IMU), (ii) LiDAR-Only, and (iii) Camera-Only.

1. Global Trajectory Consistency (ATE)

Table I presents the global drift accumulated and the Absolute Trajectory Error (ATE) by the robot during the warehouse traversal.

Sensor Configuration	**Mean ATE (m)**	**RMSE ATE (m)**	**Std Dev (m)**
Multi-Modal Fusion	0.031	0.034	0.012
LiDAR-Only	0.097	0.103	0.031
Camera-Only	0.289	0.341	0.178

Table 1: Absolute Trajectory Error across sensor configurations. All values in metres

The Multi-Modal Fusion configuration achieved the lowest mean ATE of 0.031 m, representing a 68% reduction in global drift compared to the LiDAR-only baseline. In autonomous logistics, a spatial error of less than 5 cm is generally required to safely navigate tight corridors and dock with charging stations; only the proposed fusion architecture met this standard. The Camera-Only configuration's massive maximum ATE (1.820 m) reflects complete mapping failure caused by the inability to recover global coordinates after the lighting blackout.

2. Local Odometry Smoothness (RPE)

RPE exposes the immediate consequences of sensor degradation. The sharp spike in the Camera-Only max RPE (0.812 m/s) corresponds directly to the exact moment the simulated lights were disabled, resulting in violent, erratic pose jumps. In contrast, the Fusion system's maximum RPE remained heavily suppressed at 0.047 m/s throughout the entire run. This proves that the EKF and LiDAR ICP scan-matching seamlessly compensated for the sudden loss of

visual odometry, preventing the robot from executing dangerous, erratic movements.

Table II isolates the short-term, frame-to-frame motion stability of the robot.

Sensor Configuration	**Mean RPE (m/s)**	**RMSE RPE (m/s)**	**Max RPE (m/s)**	**Std Dev (m/s)**
Multi-Modal Fusion	**0.018**	***0.027***	***0.041***	***0.009***
LiDAR-Only	***0.044***	***0.051***	***0.138***	***0.024***
Camera-Only	***0.173***	***0.219***	***0.812***	***0.142***

Table 2: Relative Pose Error across sensor configurations. All values in meters per second.

3. Graceful Degradation During Sensor Blackout

To definitively quantify the Proxy Architecture's ability to survive sudden hardware or environmental failures, a time-segmented analysis was performed exclusively on the fusion system during the blackout test.

Phase	**Active Modality**	**Mean ATE (m)**	**Mean RPE (m/s)**
Pre-Blackout (0–45s)	LiDAR + RGB-D	0.024	*0.014*
Blackout Transition	ICP fallback	0.038	0.047
During Blackout (45–110s)	LiDAR (ICP)	0.039	0.022
Overall Degradation		Δ +0.015 m	Δ +0.008 m/s

Table 3: Time-segmented metric breakdown isolating the moment of visual failure.

This table provides the ultimate proof of the project's hypothesis. The transition from full sensor fusion to the ICP-only fallback caused a remarkably minimal degradation in performance. Across the sudden and total loss of visual data, the mean ATE increased by only 0.015 m, and the mean RPE increased by just 0.008 m/s. The system successfully maintained its trajectory with zero map fragmentation, proving that the multi-modal architecture is qualitatively transformative and capable of graceful degradation.

- Map Completeness: Measures the ability to explore and map the entire bounded environment without catastrophic tracking loss.
- Trajectory Stability: Evaluates the smoothness of the estimated odometry path (blue lines) against simulated wheel slip.
- Loop Closure Efficiency: Evaluates the Graph-SLAM node's ability to find accurate spatial matches without oversaturating the optimizer (comparing the dense yellow web of LiDAR-only vs. the clean links of the Fused approach).
- Resource Optimization: Assesses computational load; the LiDAR-only approach scores low due to the severe density of its ICP scan-matching graph.

F. Discussion of Findings

The experimental results highlight three key observations:

1. Single Point of Failure in Visual SLAM
   The camera-only configuration demonstrates clear instability, particularly in low-light conditions. The failure to extract reliable visual features results in tracking loss and incomplete mapping, confirming that visual SLAM alone is insufficient for robust warehouse navigation.
2. Geometric Ambiguity in LiDAR SLAM
   While the LiDAR-only configuration successfully reconstructs the environment geometry, the loop closure graph reveals excessive connections. This indicates difficulty in distinguishing repetitive structures such as warehouse aisles, leading to redundant constraints and increased computational load.
3. Optimal Sensor Fusion Performance
   The fusion-based architecture effectively combines the strengths of both sensing modalities. LiDAR provides robust geometric constraints, while the depth camera contributes local detail and improved loop closure validation. This results in a clean and optimized graph structure, stable trajectory estimation, and complete environment mapping.

VII. Challenges Encountered and Solved

During the research and development phase, several difficulties were systematically addressed:

A. Sensor Time Synchronization.

In multi-modal SLAM systems, algorithms such as RTAB-Map construct a graph where each node represents the robot's state at a specific time instance. For accurate node creation, all sensor measurements (e.g., RGB image, depth map, and LiDAR scan) must be temporally aligned. This alignment ensures that visual features correspond correctly to geometric measurements in the environment.

However, the RGB-D camera operates at 30 Hz, while the 2D LiDAR operates at 10 Hz. As a result, sensor messages arrive at different time intervals within the ROS 2 execution pipeline. If RTAB-Map receives a camera frame without a corresponding LiDAR scan with a matching timestamp, the data is discarded to prevent incorrect data association. This leads to frequent frame drops, resulting in incomplete maps and fragmented trajectories.

To address this issue, the ROS 2 message filters package was employed using an Approximate Time synchronization policy. Instead of directly forwarding incoming sensor messages, both /image_raw and /scan topics are buffered in synchronized queues.

The synchronizer evaluates the timestamps of incoming messages and applies a tolerance window (referred to as slop), allowing messages with slightly mismatched timestamps to be paired. When a valid match is found within this window, the messages are combined and passed to RTAB-Map through a single synchronized callback.

This approach effectively compensates for differing sensor frequencies and clock inconsistencies, ensuring consistent multi-modal data fusion. As a result, frame drops are significantly reduced, leading to smoother trajectories and more complete occupancy maps.

B. Odometry Drift during Rotation

Wheel odometry estimates robot motion using encoder-based kinematic calculations. In a differential drive robot, in-place rotation is achieved by driving the wheels in opposite directions. However, due to wheel slip and frictional effects—especially evident in realistic Gazebo simulations—the actual rotation of the robot is often less than what is estimated by the encoders.

Since odometry is computed through time integration, even small angular errors accumulate rapidly. For example, if the robot estimates a 90° rotation but physically achieves only 85°, all subsequent motion will be projected along an incorrect heading, resulting in significant positional drift in the global map.

To mitigate this issue, the robot localization package was used to implement an Extended Kalman Filter (EKF) for sensor fusion. The EKF combines measurements from multiple sensors while accounting for their respective uncertainties through covariance modelling.

Wheel encoders provide reliable linear velocity estimates during straight motion but are less accurate for rotational motion due to slippage. In contrast, the IMU gyroscope directly measures angular velocity ($\dot{\theta}$) and is largely unaffected by ground interaction, making it more reliable for rotational estimation.

The EKF maintains a state vector representing the robot's pose and velocity:

$$x = [x \; y \; \theta \; \dot{x} \; \dot{y} \; \dot{\theta}]^T$$

where (x, y) denote position, (θ) represents yaw, and the dotted terms correspond to linear and angular velocities.

During the prediction step, the EKF propagates the state using encoder-derived motion estimates. In the update step, IMU measurements are incorporated to correct the predicted state. By assigning higher covariance (lower confidence) to encoder-based rotational estimates and lower covariance (higher confidence) to IMU angular measurements, the filter effectively compensates for wheel slip.

In the ROS 2 implementation, this fused estimate is published as the transformation between the `odom` and `base_link` frames, ensuring consistency within the TF tree.

The resulting filtered `/odom` topic provides a significantly more stable estimate of the robot's orientation, reducing rotational drift and improving trajectory accuracy. This stabilization is critical for downstream SLAM processes, as it provides a reliable motion prior for RTAB-Map's graph optimization and loop closure detection.

VIII. Conclusion

This work presented a ROS 2-based SLAM framework for a differential drive mobile robot operating in a simulated warehouse environment. The study focused on evaluating the robustness of different sensor configurations, including camera only, LiDAR only, and a multi-modal fusion approach implemented using RTAB-Map.

Through a series of controlled experiments, it was demonstrated that single sensor SLAM systems exhibit critical limitations. The camera-only configuration failed to maintain localization under low light and feature poor conditions, leading to early tracking loss and incomplete mapping. In contrast, the LiDAR only configuration successfully reconstructed the environment but produced an excessively dense loop closure graph, indicating geometric ambiguity and increased computational load in structured environments.

The proposed multi-modal fusion architecture effectively addressed these limitations by combining the complementary strengths of both sensing modalities. LiDAR provided stable long range geometric constraints, while the RGB-D camera contributed dense local information for improved motion estimation. This resulted in accurate map reconstruction, stable trajectory estimation, and efficient loop closure optimization, even under challenging conditions such as visual degradation.

Overall, the results confirm that sensor fusion is essential for achieving robust and reliable SLAM performance in real world warehouse scenarios. The fusion based approach eliminates the single point of failure inherent in visual SLAM and improves both mapping quality and system stability.

Future work will focus on extending the system toward higher-level autonomy by integrating semantic perception capabilities, such as object detection using YOLO based models, enabling the robot to not only map the environment but also understand and interact with it.

References

[1] R. Mur-Artal and J. D. Tardós, "ORB-SLAM2: An open-source SLAM system for monocular, stereo, and RGB-D cameras," IEEE Transactions on Robotics, vol. 33, no. 5, pp. 1255–1262, Oct. 2017, doi: 10.1109/TRO.2017.2705103.

[2] W. Hess, D. Kohler, H. Rapp, and D. Andor, "Real-Time Loop Closure in 2D LIDAR SLAM," in Proc. IEEE Int. Conf. Robotics and Automation (ICRA), Stockholm, Sweden, 2016, pp. 1271–1278.

[3] M. Labbé and F. Michaud, "Online Global Loop Closure Detection for Large-Scale Multi-Session Graph-Based SLAM," IEEE Transactions on Robotics, vol. 29, no. 6, pp. 1480–1497, Dec. 2013.

[4] G. Grisetti, C. Stachniss, and W. Burgard, "A tutorial on graph-based SLAM," IEEE Intelligent Transportation Systems Magazine, vol. 2, no. 4, pp. 31–43, 2010.

[5] Black Coffee Robotics, *BCR Bot*, GitHub, 2023. https://github.com/blackcoffeerobotics/bcr_bot

[6] S. Macenski et al., "SLAM Toolbox: SLAM for the dynamic world," *Journal of Open Source Software*, vol. 6, no. 61, 2021.

[7] T. Moore and D. Stouch, "A Generalized Extended Kalman Filter Implementation for the Robot Operating System," in Proc. IEEE Int. Conf. on Intelligent Autonomous Systems (IAS), 2016.

[8] M. Labbé and F. Michaud, "RTAB-Map as an Open-Source Lidar and Visual SLAM Library for Large-Scale and Long-Term Online Operation," *Journal of Field Robotics*, vol. 36, no. 2, pp. 416–446, 2019.